

Homework 3

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1 Mini-batch Stochastic Gradient Descent

The mini-batch stochastic gradient descent (SGD) method aims to solve the following stochastic optimization problem:

$$\min_{x \in \mathbb{R}^d} f(x) \triangleq \mathbb{E}_{\xi \sim \mathcal{D}} [F(x; \xi)],$$

We consider the mini-batch SGD algorithm described in Algorithm 1:

Algorithm 1 Mini-batch SGD

Require: Objective function F , initial point x_0 , data distribution $\mathcal{D}$, step size γ , batch size B

- 1: **for** $k = 0, 1, 2, \dots, K - 1$ **do**
- 2: Sample i.i.d. mini-batch: $\xi_{(k,1)}, \dots, \xi_{(k,B)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{D}$
- 3: Compute stochastic gradient: $g_k = \frac{1}{B} \sum_{b=1}^B \nabla F(x_k; \xi_{(k,b)})$
- 4: Update parameters: $x_{k+1} = x_k - \gamma \cdot g_k$
- 5: **end for**

Ensure: Parameter sequence $\{x_k\}_{k=0}^K$

To analyze the convergence properties of mini-batch SGD, we make the following assumptions:

1. The objective function f is L -smooth.
2. Let $\mathcal{F}_k = \{\xi_{(0,1)}, \dots, \xi_{(0,B)}, x_1, \xi_{(1,1)}, \dots, \xi_{(1,B)}, \dots, x_{k-1}, \xi_{(k-1,1)}, \dots, \xi_{(k-1,B)}, x_k\}$ denote the filtration containing all randomness up to iteration k . For any $k \leq K - 1$ and $1 \leq b \leq B$, the stochastic gradient $\nabla F(x_k; \xi_{(k,b)})$ satisfies:

$$\begin{aligned} \mathbb{E} [\nabla F(x_k; \xi_{(k,b)}) \mid \mathcal{F}_k] &= \nabla f(x_k), \\ \mathbb{E} [\|\nabla F(x_k; \xi_{(k,b)}) - \nabla f(x_k)\|_2^2 \mid \mathcal{F}_k] &\leq \sigma^2. \end{aligned}$$

Based on these assumptions, prove the following:

1. For any $0 \leq k \leq K - 1$, show that:

$$\mathbb{E} [\|g_k - \nabla f(x_k)\|_2^2 \mid \mathcal{F}_k] \leq \frac{\sigma^2}{B}.$$

2. If the step size satisfies $\gamma \leq \frac{1}{L}$, prove that:

$$\frac{1}{K+1} \sum_{k=0}^K \mathbb{E} [\|\nabla f(x_k)\|_2^2] \leq \frac{2(f(x_0) - f^*)}{\gamma(K+1)} + \frac{L\gamma\sigma^2}{B},$$

where $f^* \triangleq \inf_{x \in \mathbb{R}^d} f(x)$.

2 The Influence of Noise Structure

Consider the minimization of the quadratic function:

$$f(x) = \frac{1}{2} \sum_{i=1}^d a_i x_i^2,$$

where $a_1 \geq a_2 \geq \dots \geq a_d > 0$. The problem is typically ill-conditioned, i.e., the condition number $\kappa := \frac{a_1}{a_d} \gg 1$. We analyze stochastic gradient descent (SGD) with a constant step size γ .

The stochastic gradient $\nabla f(x; \xi)$ is an unbiased estimator of the true gradient $\nabla f(x)$, and its noise covariance is defined as:

$$\Sigma(x) = \mathbb{E} \left[(\nabla f(x; \xi) - \nabla f(x)) (\nabla f(x; \xi) - \nabla f(x))^\top \mid x \right].$$

We compare the following three noise structures:

- (i) **Isotropic noise:** $\Sigma(x) = \sigma^2 I$.
- (ii) **Hessian-scaled noise:** $\Sigma(x) = \sigma^2 \text{diag}(a_1, \dots, a_d)$.
- (iii) **Mini-batch SGD noise:** $\nabla f(x; \xi) = a_i x_i$, where index i is sampled uniformly from $\{1, 2, \dots, d\}$.

Questions:

1. Starting from an arbitrary non-zero initial point x_0 , and assuming $\gamma < \frac{2}{a_1}$:

- (a) Derive the recurrence relation between $\mathbb{E}[x_{k+1}]$ and $\mathbb{E}[x_k]$, and show that for all three noise structures,

$$\lim_{k \rightarrow \infty} \mathbb{E}[x_k] = 0.$$

- (b) Derive a recurrence relation (or inequality) between $\mathbb{E}[\|x_{k+1}\|^2]$ and $\mathbb{E}[\|x_k\|^2]$.

- For isotropic and Hessian-scaled noise, provide an upper bound for

$$\limsup_{k \rightarrow \infty} \mathbb{E}[\|x_k\|^2].$$

- For the mini-batch SGD noise, determine an additional condition on γ that ensures $\mathbb{E}[\|x_k\|^2] \rightarrow 0$ as $k \rightarrow \infty$.

2. Discuss how the structure and magnitude of the noise covariance matrix affect the convergence.